

UIT_HighDefinition at ImageCLEFmedical Caption 2026: CUI-Guided Qwen3-VL for Medical Image Caption Prediction

Notebook for the DS-MED Lab (UIT) at CLEF 2026

Nguyen Tan Duy^{1,2}, Nguyen Thi Quynh Han^{1,2}, Nguyen Minh Triet^{1,2} and Thien B. Nguyen-Tat^{1,2,*}

¹University of Information Technology, Ho Chi Minh City, Vietnam

²Vietnam National University, Ho Chi Minh City, Vietnam

Abstract

Purpose: This paper presents the UIT_HighDefinition team's system for the ImageCLEFmedical Caption 2026 Caption Prediction task. The task requires generating concise and clinically meaningful captions for biomedical images from heterogeneous medical modalities. **Method:** We propose a CUI-guided vision-language captioning pipeline based on Qwen3-VL-8B. Instead of relying only on the visual input, our method incorporates predicted UMLS Concept Unique Identifiers (CUIs) as soft clinical hints in the prompt. The generated captions are refined using rule-based post-processing to remove generic prefixes, normalize modality terms, control caption length, and ensure valid submission formatting. To process the full test set efficiently under Kaggle runtime constraints, we distribute inference across three machines and six GPU workers, with checkpoint-based resume support. **Results:** Our final submission achieved an official overall score of 0.3193. The system obtained a semantic similarity score of 0.8496, BERT score of 0.5977, ROUGE score of 0.2243, BLEURT score of 0.3278, MedCAT score of 0.1397, and alignment score of 0.1379. **Conclusion:** The results suggest that large vision-language models can generate useful medical image captions when guided by domain-specific concept hints. However, the relatively lower MedCAT and alignment scores indicate that exact biomedical concept grounding remains challenging.

Keywords

Medical image captioning, ImageCLEFmedical Caption, Vision-language model, Qwen3-VL, UMLS, CUI-guided prompting, Medical image understanding

1. Introduction

Medical image captioning aims to automatically generate textual descriptions that summarize the visual and clinical content of biomedical images. This task is important because medical images are widely used in scientific articles, clinical documentation, education, and diagnostic workflows. A reliable captioning system can support physicians and researchers by producing concise descriptions of imaging modalities, anatomical regions, abnormalities, and relevant visual findings. Nevertheless, medical image captioning remains difficult because biomedical images are highly diverse, covering modalities such as X-ray, computed tomography (CT), magnetic resonance imaging (MRI), ultrasound, endoscopy, histology, microscopy, angiography, and hybrid imaging techniques.

Compared with general-domain image captioning, medical caption generation requires stronger domain knowledge and more careful language control. The output must be clinically relevant, concise, and grounded in visual evidence. A generic caption such as “a medical image” is not sufficient; the system is expected to identify the modality, anatomical region, and visible finding when possible. At the same time, hallucinating unsupported diseases or patient information is particularly problematic in the medical domain. These challenges motivate the use of large vision-language models (VLMs) combined with domain-specific guidance.

CLEF 2026 Working Notes, 9–12 September 2026, Evora, Portugal

*Corresponding author.

✉ 24520389@gm.uit.edu.vn (N. T. Duy); 24520013@gm.uit.edu.vn (N. T. Q. Han); 24521852@gm.uit.edu.vn (N. M. Triet); thienntb@uit.edu.vn (T. B. Nguyen-Tat)



© 2026 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

ImageCLEFmedical Caption is a long-running benchmark for medical image understanding. Recent editions have included concept detection, caption prediction, and explainability tasks. Previous top-performing systems have explored encoder-decoder models, BLIP-based architectures, InstructBLIP, retrieval-augmented generation, concept-aware generation, reranking, and ensemble strategies. These systems show that medical image captioning benefits from strong multimodal backbones, domain-specific concepts, and careful post-processing.

In this paper, we describe the UIT_HighDefinition team’s system for the ImageCLEFmedical Caption 2026 Caption Prediction task. Our method uses Qwen3-VL-8B as the main caption generation model. To inject medical domain information, we convert predicted UMLS Concept Unique Identifiers (CUIs) into textual clinical hints and include them in the prompt. Unlike hard-label conditioning, these CUI terms are treated as soft hints: the model is explicitly instructed to use them only when they are visually supported by the image. This design aims to improve medical terminology coverage while reducing the risk of unsupported clinical claims.

The main contributions of our work are summarized as follows:

- We propose a Qwen3-VL-8B-based medical image captioning pipeline for ImageCLEFmedical Caption 2026.
- We introduce CUI-guided prompting, where predicted UMLS concepts are used as soft clinical hints rather than mandatory labels.
- We design a rule-based caption post-processing module to remove generic phrases, normalize modality terms, control length, and ensure clean outputs.
- We implement distributed inference across three Kaggle machines and six GPU workers, with a resume mechanism to handle runtime interruptions.
- We report official evaluation results and analyze strengths and limitations of the proposed system.

The remainder of this paper is organized as follows. Section 2 discusses related work. Section 3 describes the task and dataset. Section 4 presents the proposed pipeline. Section 5 reports the experimental setup and official results. Section 6 discusses limitations and observations. Section 7 concludes the paper.

2. Background and Related Work

2.1. Medical image captioning

Medical image captioning combines computer vision and natural language generation. The task requires mapping visual information from biomedical images into clinically meaningful text. Early systems often relied on encoder-decoder architectures, where a visual encoder extracted image features and a language decoder generated captions. With the development of transformer architectures and large-scale vision-language pretraining, recent systems have increasingly adopted multimodal foundation models.

In ImageCLEFmedical Caption 2023, top-performing approaches included encoder-decoder models, BLIP-2-based systems, caption denoising, and retrieval-based strategies. The VCMi team explored both concept detection and caption prediction using multi-label classification, retrieval, and language generation approaches, showing that language generation methods with vision encoders and language decoders were competitive for caption prediction. In ImageCLEFmedical Caption 2024, several systems investigated transformer-based architectures, including BLIP, BioBART-based captioning, and Query Transformer modules. These works demonstrate a gradual shift from task-specific captioning architectures toward stronger vision-language models.

2.2. Vision-language models for medical images

Vision-language models have become an important direction for medical image captioning. BLIP and InstructBLIP have been widely used because they can align image and language representations

and transfer to image-to-text tasks. In the 2025 ImageCLEFmedical Caption challenge, several high-performing teams used InstructBLIP or other large VLMs as caption generation backbones. These systems often combined initial caption generation with additional refinement, retrieval, reranking, or fusion modules.

Although large VLMs provide strong general visual-language understanding, they may still produce captions that are too generic or insufficiently grounded in medical terminology. This limitation is especially visible when the image contains specialized findings or uncommon anatomical structures. Therefore, several approaches have attempted to incorporate biomedical knowledge, predicted concepts, or retrieval examples to improve clinical specificity.

2.3. Concept-aware caption generation

Concept detection is closely related to medical caption generation. In ImageCLEFmedical Caption, concept detection aims to associate biomedical images with UMLS concepts, while caption prediction aims to generate descriptive captions. Prior work has shown that concept information can improve caption generation by providing medically relevant terms that should be considered by the generation model. For example, concept-aware systems have integrated predicted concepts into attention mechanisms, prompt construction, or downstream caption refinement.

However, concept-guided generation also introduces risks. If predicted CUIs are noisy, incomplete, or visually unsupported, forcing the captioning model to include them may degrade the output and increase hallucination. Retrieval-augmented approaches face a similar problem: retrieved examples can be useful only when they are highly relevant, while noisy retrievals can mislead the generator. Our method follows a soft-guidance strategy. CUI terms are provided to Qwen3-VL as possible clinical hints, but the prompt explicitly instructs the model not to force unsupported diseases or findings.

3. Task and Dataset Description

3.1. Caption Prediction task

The ImageCLEFmedical Caption 2026 Caption Prediction task requires participants to generate one caption for each test image. The submitted file must contain exactly two columns, ID and Caption. Each row corresponds to one official test image. The generated caption should describe the visible medical content of the image, including the modality, anatomical region, and main visual finding when these elements are identifiable.

The evaluation uses multiple metrics that capture complementary aspects of caption quality. Textual similarity metrics assess the semantic and lexical closeness between generated captions and reference captions, while clinical grounding metrics evaluate the coverage and alignment of biomedical concepts. This multi-metric evaluation reflects the fact that medical captioning requires both fluent language generation and accurate clinical content.

3.2. Dataset characteristics

The official test set used in our experiments contains 15,249 medical images. The images are drawn from biomedical literature and cover a broad range of imaging modalities and visual styles. Unlike natural image captioning datasets, the images may contain radiographs, cross-sectional scans, endoscopic views, microscopic images, diagrams, plots, and composite figures. This diversity makes it difficult to rely on a single modality-specific assumption.

For our pipeline, each image is identified by its official ID. In addition to the image input, we use predicted CUIs from a separate concept detection submission as optional clinical hints. These CUIs are mapped to textual terms and incorporated into the prompt only as auxiliary information.

4. Proposed Method

4.1. Overall pipeline

Figure 1 summarizes our proposed pipeline. The input consists of a test image and, when available, a list of predicted CUIs. The CUIs are converted into readable clinical terms. The image and the CUI-derived hints are then passed to Qwen3-VL-8B through an instruction prompt. The generated caption is cleaned by a rule-based post-processing module. Finally, all captions are merged into a valid `submission.csv` file.

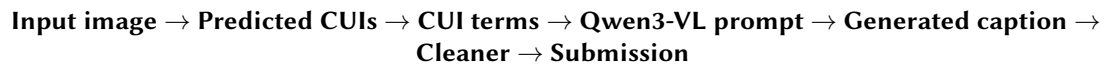


Figure 1: Overview of the proposed CUI-guided Qwen3-VL caption generation pipeline.

The design focuses on four practical goals. First, the model should produce clinically informative captions rather than generic descriptions. Second, the prompt should encourage visual grounding and avoid forcing unsupported diagnoses. Third, the output must be robustly formatted for strict submission checking. Fourth, the inference process must be efficient enough to handle more than fifteen thousand images under limited GPU runtime.

4.2. CUI-guided prompt construction

We use predicted UMLS CUIs as domain-specific hints. Each CUI is mapped to a textual medical term using a CUI-to-metadata dictionary. To avoid overloading the prompt, we keep only a limited number of unique terms for each image. These terms are inserted into the instruction as possible clinical hints.

The prompt contains two important constraints. First, it asks the model to focus on visible medical content and mention the modality when clear. Second, it explicitly states that the CUI hints should be used only if they are visually supported by the image. This is important because the concept prediction stage may introduce noisy terms. By using the CUIs as soft guidance, the model can benefit from domain-specific vocabulary without being forced to include every predicted concept.

A simplified version of the instruction is shown below:

Write one concise, clinically relevant English caption for the image. Mention modality when clear: CT, MRI, X-ray, ultrasound, histology, endoscopy, angiography, PET/CT, or microscopy. Possible UMLS/CUI clinical hints are provided. Use these hints only if they are visually supported by the image. Do not force unsupported diseases or findings. Output only one caption sentence.

4.3. Caption generation with Qwen3-VL

We use Qwen3-VL-8B-Instruct as the main caption generation model. For each test sample, the model receives the image and the text prompt described above. Generation is performed deterministically without sampling to improve reproducibility. We set the maximum number of generated tokens to 80, which is sufficient for a concise caption while reducing the chance of overly long outputs.

Because Kaggle T4 GPUs have limited memory, the model is loaded with 4-bit quantization. We also constrain the visual input resolution using minimum and maximum pixel limits. This enables inference on large numbers of images without exceeding memory limits.

4.4. Caption post-processing

The raw model outputs are passed through a rule-based cleaning module. This module performs the following operations:

- removing unwanted prefixes such as “Caption:”, “Final caption:”, “This image shows”, and similar phrases;

- normalizing modality expressions, for example replacing “computed tomography” with “CT” and “magnetic resonance imaging” with “MRI”;
- removing excessive whitespace and line breaks;
- trimming overly long captions while preserving complete sentence boundaries when possible;
- ensuring that each caption ends with proper punctuation;
- replacing empty or invalid captions with a fallback caption.

This cleaning step is simple but important. The official submission checker is strict about file format, and generic phrases can reduce caption quality. The cleaner therefore improves both technical validity and linguistic consistency.

4.5. Distributed inference and resume strategy

The full test set contains 15,249 images. To process it efficiently, we divide the images into six global shards using their sorted IDs. We use three Kaggle machines, each with two NVIDIA Tesla T4 GPUs. Each machine runs two workers, and each worker processes one shard. The six workers correspond to ranks 0 to 5.

Runtime interruptions are common in cloud notebook environments. To prevent losing progress, each worker periodically saves its intermediate results to a CSV file. When a machine times out, the corresponding rank can be resumed by loading the previous CSV, skipping already processed IDs, and generating captions only for missing images. This mechanism allowed us to complete the full inference without restarting finished shards.

5. Experiments and Results

5.1. Implementation details

All inference experiments were conducted on Kaggle using NVIDIA Tesla T4 GPUs. The final inference used three machines and six workers. Qwen3-VL-8B-Instruct was loaded with 4-bit quantization to fit within GPU memory. The maximum output length was set to 80 new tokens. The visual input size was controlled by pixel constraints to balance visual detail and inference speed.

After all shards were completed, the six CSV files were merged into a single submission file. We verified that the final submission contained 15,249 rows, no duplicate IDs, no empty captions, and the exact required header `ID, Caption`.

5.2. Official results

Table 1 reports the official scores of our final submission.

Table 1

Official results of the UIT_HighDefinition submission on ImageCLEFmedical Caption 2026.

Metric	Score
Overall	0.3193
Score	0.4999
Score secondary	0.1388
BERT	0.5977
ROUGE	0.2243
Similarity	0.8496
BLEURT	0.3278
MedCAT	0.1397
Align	0.1379

The overall score of our system is 0.3193. The high similarity score of 0.8496 indicates that the generated captions are generally semantically close to the references. The BERT and BLEURT scores further suggest that the captions have reasonable language-level relevance. However, the MedCAT and alignment scores are lower, showing that exact biomedical concept coverage and factual grounding remain challenging.

5.3. Output analysis

We performed additional validation on the final submission. The file contained 15,249 captions, with no duplicate IDs and no empty outputs. The number of unique captions was 13,666, indicating that the model did not collapse into a small set of repeated generic captions. Our cleaning module also removed common undesirable phrases such as “This image shows”, “Caption:”, and overly generic outputs.

Qualitatively, many captions correctly identified the imaging modality and anatomical region. The model was often able to generate concise descriptions for radiographs, CT scans, MRI images, histology slides, and endoscopic images. Nevertheless, errors remained in difficult cases, especially when images were ambiguous, contained multiple panels, or required specialized clinical knowledge not directly visible from the image.

6. Discussion

The results suggest that Qwen3-VL-8B can be a strong backbone for medical image caption prediction when combined with task-specific prompting and post-processing. The use of CUI-derived hints provides additional biomedical context, which can help the model select more appropriate terminology. Compared with using the image alone, concept hints can guide the model toward clinically relevant descriptions.

At the same time, the system has several limitations. First, the quality of the final caption depends on the quality of the predicted CUIs. Incorrect or incomplete CUI hints may mislead the model. Although our prompt treats CUIs as soft guidance, hallucination cannot be completely eliminated. Second, the method does not fine-tune Qwen3-VL on the ImageCLEFmedical dataset. As a result, it relies on the model’s pretrained knowledge and prompt-following ability. Third, rule-based post-processing improves consistency but cannot verify whether a generated medical statement is factually correct.

The lower MedCAT and alignment scores suggest that future work should focus on better concept grounding. Possible directions include using a stronger concept detection model, generating multiple candidate captions followed by medical reranking, integrating MedCLIP or BioMedCLIP-based visual-textual scoring, and fine-tuning a lightweight caption refiner on validation data. Another promising direction is to use explainability methods to connect generated medical terms to image regions, improving trust and reducing unsupported claims.

7. Conclusion

This paper presented the UIT_HighDefinition team’s system for the ImageCLEFmedical Caption 2026 Caption Prediction task. Our approach uses Qwen3-VL-8B with CUI-guided prompting, rule-based caption cleaning, and distributed inference with checkpoint-based resume support. The final submission achieved an official overall score of 0.3193.

The study shows that large vision-language models can generate useful medical image captions when guided by domain-specific concept hints. However, the gap between semantic similarity and clinical grounding metrics indicates that medical concept accuracy remains a key challenge. Future work will explore improved concept prediction, candidate reranking, and stronger visual grounding mechanisms.

Acknowledgments

This work was conducted as part of the UIT_HighDefinition team’s participation in the ImageCLEFmedical Caption 2026 challenge. We thank the ImageCLEF organizers for providing the benchmark and evaluation platform.

Declaration on Generative AI

During the preparation of this work, the authors used generative AI tools to support grammar checking, wording refinement, and LaTeX formatting. The system implementation, experiments, interpretation of results, and final manuscript were reviewed and edited by the authors, who take full responsibility for the content of this publication.

References

- [1] S. Bai, Y. Cai, R. Chen, K. Chen, X. Chen, Z. Cheng, L. Deng, W. Ding, C. Gao, C. Ge, et al., Qwen3-VL technical report, arXiv preprint arXiv:2511.21631 (2025). URL: <https://arxiv.org/abs/2511.21631>.
- [2] J. Rückert, L. Bloch, R. Brüngel, A. Idrissi-Yaghir, H. Schäfer, C. S. Schmidt, S. Koitka, O. Pelka, A. Ben Abacha, A. García Seco de Herrera, H. Müller, P. A. Horn, F. Nensa, C. M. Friedrich, ROCov2: Radiology objects in COnText version 2, an updated multimodal image dataset, *Scientific Data* 11 (2024) 688. URL: <https://doi.org/10.1038/s41597-024-03496-6>. doi:10.1038/s41597-024-03496-6.
- [3] J. Rückert, L. Bloch, A. Idrissi-Yaghir, R. Brüngel, H. Schäfer, S. Koitka, H. Müller, C. M. Friedrich, Overview of ImageCLEFmedical 2023 – caption prediction and concept detection, in: *CLEF 2023 Working Notes*, volume 3497 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2023. URL: <https://ceur-ws.org/Vol-3497/paper-108.pdf>.
- [4] J. Rückert, H. Damm, A. Idrissi-Yaghir, R. Brüngel, L. Bloch, H. Schäfer, S. Koitka, H. Müller, C. M. Friedrich, Overview of ImageCLEFmedical 2024 – caption prediction and concept detection, in: *CLEF 2024 Working Notes*, volume 3740 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2024. URL: <https://ceur-ws.org/Vol-3740/paper-132.pdf>.
- [5] H. Damm, J. Rückert, A. Ben Abacha, A. García Seco de Herrera, H. Müller, et al., Overview of ImageCLEFmedical 2025 – medical concept detection and interpretable caption generation, in: *CLEF 2025 Working Notes*, volume 4038 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2025. URL: https://ceur-ws.org/Vol-4038/paper_170.pdf.
- [6] R. Pan, T. Bernal-Beltrán, J. A. García-Díaz, R. Valencia-García, UMUTeam at ImageCLEF 2025: Fine-tuning a vision-language model for medical image captioning and SapBERT-based reranking for concept detection, in: *CLEF 2025 Working Notes*, volume 4038 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2025, pp. 2553–2563. URL: https://ceur-ws.org/Vol-4038/paper_200.pdf.
- [7] J. He, S. Ferdowsi, W. Feng, F. Alves, A. Platon, D. Teodoro, DS4DH group at ImageCLEFmedical caption 2025, in: *CLEF 2025 Working Notes*, volume 4038 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2025, pp. 2448–2460. URL: <https://archive-ouverte.unige.ch/unige:192276>.
- [8] A. Chatzipapadopoulou, I. Pantelidis, F. Charalampakos, M. Samprovalaki, G. Moschovis, P. Kaliosis, K. V. Dalakleidi, J. Pavlopoulos, I. Androutsopoulos, AUEB NLP Group at ImageCLEFmedical caption 2025, in: *CLEF 2025 Working Notes*, volume 4038 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2025.
- [9] J. Angulo, J. Aguilar, JJ-VMed: A framework for automated concepts, captions and explainability of medical image, in: *CLEF 2025 Working Notes*, volume 4038 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2025. URL: https://ceur-ws.org/Vol-4038/paper_179.pdf.
- [10] B. Yang, Y. Yu, Y. Zou, T. Zhang, PCLmed: Champion solution for ImageCLEFmedical 2024 caption prediction challenge via medical vision-language foundation models, in: *CLEF 2024*

Working Notes, volume 3740 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2024. URL: <https://ceur-ws.org/Vol-3740/paper-164.pdf>.

- [11] Q. V. Nguyen, H. Q. Pham, D. Q. Tran, T. K.-B. Nguyen, N.-H. Nguyen-Dang, T. B. Nguyen-Tat, UIT-DarkCow team at ImageCLEFmedical caption 2024: Diagnostic captioning for radiology images efficiency with transformer models, in: *CLEF 2024 Working Notes*, volume 3740 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2024. URL: <https://ceur-ws.org/Vol-3740/paper-158.pdf>.
- [12] N. N.-Y. Nguyen, H. L. Tu, P. D. Nguyen, T. N. Do, T. M. Thai, T. B. Nguyen-Tat, DS@BioMed at ImageCLEFmedical caption 2024: Enhanced attention mechanisms in medical caption generation through concept detection integration, in: *CLEF 2024 Working Notes*, volume 3740 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2024. URL: <https://arxiv.org/abs/2406.00391>.
- [13] T. V. Phan, T. K. Nguyen, Q. A. D. D. Hoang, Q. T. Phan, T. B. Nguyen-Tat, UIT-2Q2T at ImageCLEFmedical 2024 caption: Multimodal medical image captioning using bootstrapping language-image pre-training, in: *CLEF 2024 Working Notes*, volume 3740 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2024.
- [14] I. Rio-Torto, C. Patrício, H. Montenegro, T. Gonçalves, J. S. Cardoso, Detecting concepts and generating captions from medical images: Contributions of the VCMi team to ImageCLEFmedical caption 2023, in: *CLEF 2023 Working Notes*, volume 3497 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2023. URL: <https://ceur-ws.org/Vol-3497/paper-137.pdf>.
- [15] A. Nicolson, J. Dowling, B. Koopman, A concise model for medical image captioning, in: *CLEF 2023 Working Notes*, volume 3497 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2023. URL: https://github.com/aehrc/imageclefmedical_caption_23.
- [16] T. T. Le, T. M. Ngo, K. D. Nguyen, T. H. Dang, T. H. Pham, T. B. Nguyen-Tat, Anastasia at MEDIQA-MAGIC 2025: A multi-approach segmentation framework with extensive augmentation, in: *CLEF 2025 Working Notes*, CEUR Workshop Proceedings, 2025.
- [17] N. P. H. Le, H. P. D. Huy, H. T. Dinh Nhat, H. T. Minh, T. B. Nguyen-Tat, DermoSegDiff and DermKEM for comprehensive dermatology ai, in: *CLEF 2025 Working Notes*, CEUR Workshop Proceedings, 2025.
- [18] J. Li, D. Li, C. Xiong, S. C. H. Hoi, BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation, in: *Proceedings of the 39th International Conference on Machine Learning*, 2022, pp. 12888–12900.
- [19] J. Li, D. Li, S. Savarese, S. C. H. Hoi, BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models, in: *Proceedings of the 40th International Conference on Machine Learning*, 2023, pp. 19730–19742.
- [20] W. Dai, J. Li, D. Li, A. M. H. Tiong, J. Zhao, W. Wang, B. Li, P. Fung, S. C. H. Hoi, Instructblip: Towards general-purpose vision-language models with instruction tuning, in: *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [21] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, Y. Artzi, BERTScore: Evaluating text generation with BERT, in: *International Conference on Learning Representations*, 2020.
- [22] C.-Y. Lin, ROUGE: A package for automatic evaluation of summaries, in: *Text Summarization Branches Out*, 2004, pp. 74–81.
- [23] T. Sellam, D. Das, A. P. Parikh, BLEURT: Learning robust metrics for text generation, in: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 7881–7892.
- [24] O. Bodenreider, The Unified Medical Language System (UMLS): Integrating biomedical terminology, *Nucleic Acids Research* 32 (2004) D267–D270. doi:10.1093/nar/gkh061.
- [25] Z. Kraljevic, T. Searle, A. Shek, L. Roguski, K. Noor, D. Bean, A. Mascio, L. Zhu, A. A. Folarin, A. Roberts, et al., MedCAT: Medical concept annotation tool, *Artificial Intelligence in Medicine* 117 (2021) 102083. doi:10.1016/j.artmed.2021.102083.
- [26] Z. Wang, Z. Wu, D. Agarwal, J. Sun, MedCLIP: Contrastive learning from unpaired medical images and text, *arXiv preprint arXiv:2210.10163* (2022). URL: <https://arxiv.org/abs/2210.10163>.
- [27] S. Zhang, Y. Xu, N. Usuyama, H. Xu, J. Bagga, R. Tinn, S. Preston, R. Rao, M. Wei, N. Valluri, et al., BioMedCLIP: A multimodal biomedical foundation model pretrained from fifteen million scientific

- image-text pairs, arXiv preprint arXiv:2303.00915 (2023). URL: <https://arxiv.org/abs/2303.00915>.
- [28] T. B. Nguyen-Tat, T. Q. T. Nguyen, H. N. Nguyen, V. M. Ngo, Enhancing brain tumor segmentation in MRI images: A hybrid approach using UNet, attention mechanisms, and transformers, *Egyptian Informatics Journal* 27 (2024) 100528. doi:10.1016/j.eij.2024.100528.
 - [29] T. B. Nguyen-Tat, T. Q. Hung, P. T. Nam, V. M. Ngo, Evaluating pre-processing and deep learning methods in medical imaging: Combined effectiveness across multiple modalities, *Alexandria Engineering Journal* 119 (2025) 558–586. doi:10.1016/j.aej.2025.01.090.
 - [30] T. B. Nguyen-Tat, H.-A. Vo, P.-S. Dang, QMaxViT-Unet+: A query-based MaxViT-Unet with edge enhancement for scribble-supervised segmentation of medical images, *Computers in Biology and Medicine* 187 (2025) 109762. doi:10.1016/j.combiomed.2025.109762.
 - [31] V. Pham, H. Nguyen, B. Pham, T. Nguyen, H. Nguyen, Robust engineering-based unified biomedical imaging framework for liver tumor segmentation, *Current Medical Imaging* 19 (2023) 37–45. doi:10.2174/1573405617666210804151024.
 - [32] T. B. Nguyen-Tat, T.-P. Nguyen-Duong, Optimizing knee osteoarthritis severity diagnostics: A GA-enhanced deep ensemble approach in medical imaging, *Ain Shams Engineering Journal* 16 (2025) 103524. doi:10.1016/j.asej.2025.103524.